

Software Design Document
E-Bike Rental System



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Summary

The E-Bike Rental System is a web based application which is specifically designed to provide efficient and user friendly platform for electric bikes. The application allows users to register, browse available bikes, reserve rides, make secure online payments, and manage ride history, and admins can manage bikes, users, bookings, and reports through a dashboard. The application follows the MVC architecture using Django web development framework and MySQL to ensure scalability, security, and maintainability. It also integrates payment and notification APIs to provide a smooth and modern rental experience that promotes eco-friendly transportation.

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1. Application Architecture

The E-Bike Rental System is designed using the MVC (Model-View-Controller) architecture provided by the Django framework. This architecture separates the system into different components to improve scalability, maintainability, and security.

The application consists of three main layers:

Layer	Description
Presentation Layer	Handles user interface and user interaction using HTML, CSS, JavaScript, and Bootstrap
Business Logic Layer	Processes booking, authentication, payments, and notifications using Django
Database Layer	Stores user, bike, booking, and payment data using MySQL

1.1 MVC Components

Models

Models handle database operations and store application data.

Views

Views display web pages and user interfaces.

Controllers

Controllers process requests, apply business logic, and communicate between views and models.

1.2 Proposed Controllers and Functions

Controller	Main Functions
Authentication Controller	registerUser(), loginUser(), logoutUser()
Bike Controller	addBike(), updateBike(), searchBike()
Booking Controller	reserveBike(), startRide(), endRide()
Payment Controller	processPayment(), generateInvoice()
Notification Controller	sendOTP(), sendBookingAlert()
Admin Controller	manageUsers(), manageBikes(), generateReports()

1.3 APIs and Libraries Used

API/Library	Purpose
JazzCash API	Online payments
EasyPaisa API	Payment processing
SMS API	OTP and alerts
Django	Backend framework
MySQL	Database management

1.4 UML Diagrams Used

The following UML diagrams are used to explain system working:

- Use Case Diagram
- Sequence Diagram
- Class Diagram
- ER Diagram

Reference of these diagram

1.5 Coding Conventions

Element	Convention
Class Names	PascalCase
Variables	camelCase
Functions	camelCase

1.6 Definitions, Acronyms, and Abbreviations

Term	Description
SDD	Software Design Document
SRS	Software Requirements Specification
GUI	Graphical User Interface
DBMS	Database Management System
API	Application Programming Interface
OTP	One Time Password
GPS	Global Positioning System
HTTPS	HyperText Transfer Protocol Secure

1.6 References

- SRS Document of E-Bike Rental System
- JAZZCASH API
- EASY PASIA API

2. Data Model Schema

A Data Model Schema is the foundational blueprint or skeleton structure that defines how data is organized, stored, and related within a database or information system. It outlines the logical layout including tables, fields (columns), data types, and constraints without containing the actual data itself.

2.1 Entities/Tables in Schema

Notification Entity

The Notification entity stores alerts and messages sent to users.

Attribute	Description
notification_id	Unique notification ID
message	Notification content
type	Notification type
status	Delivery status
user_id	Linked user

User Table

Field	Type	Description
user_id	INT (PK)	Unique user ID
name	VARCHAR	User name
email	VARCHAR	User email
phone	VARCHAR	User phone
password	VARCHAR	Encrypted password

Bike Table

Field	Type	Description
bike_id	INT (PK)	Unique bike ID
status	VARCHAR	Availability status
battery_level	INT	Battery percentage
location	VARCHAR	Current location

Ride Table

Field	Type	Description
ride_id	INT (PK)	Ride identifier
start_time	DATETIME	Ride start
end_time	DATETIME	Ride end
distance	FLOAT	Distance traveled
cost	FLOAT	Ride cost
user_id	INT (FK)	User reference
bike_id	INT (FK)	Bike reference

Payment Table

Field	Type	Description
payment_id	INT (PK)	Payment ID
amount	FLOAT	Payment amount
payment_method	VARCHAR	Method used
status	VARCHAR	Payment status
ride_id	INT (FK)	Ride reference

Admin Table

Field	Type	Description
admin_id	INT (PK)	Admin identifier
name	VARCHAR	Admin name
email	VARCHAR	Admin email

2.2 ER Diagram Explanation

The Entity Relationship (ER) Diagram represents the structure of the database and the relationship between entities.

Main Relationships

User and Ride

A user can perform multiple rides, but each ride belongs to only one user.

Bike and Ride

A bike can be rented multiple times, but each ride uses one bike.

Ride and Payment

Each completed ride generates one payment transaction.

User and Notification

Users receive multiple system notifications.

Admin and Bike

Admins manage bike inventory and bike status.

Relationship	Type
User → Ride	One-to-Many
Bike → Ride	One-to-Many
Ride → Payment	One-to-One
User → Notification	One-to-Many
Admin → Bike	One-to-Many

2.3 Database Constraints

The database includes constraints to maintain data integrity.

Constraints Used

Constraint	Purpose
Primary Key	Unique identification
Foreign Key	Maintain relationships
NOT NULL	Prevent empty fields
UNIQUE	Prevent duplicate emails
CHECK	Validate data values

2.4 Future Database Enhancements

Future versions of the database may include:

- GPS tracking tables
- Smart lock management
- Wallet system
- AI-based ride analytics
- Multi-language support

3. User Interface

Overall design philosophy such as color scheme, font, design niche and screen constraints (if required then responsiveness or targeted screens size) will be discussed here. List of screens/pages will be provided and explain the data on each page/screen. List of controls (Textbox, Drop Down, Buttons etc.) on each page will be provided and explain the working of data. Also explain the input constraints.

3.1 User Interface Design Philosophy

The E-Bike Rental System follows a modern, clean, and responsive user interface design to provide smooth user experience for both customers and administrators. The interface is designed according to Human Computer Interaction (HCI) principles to ensure simplicity, accessibility, readability, and ease of navigation.

The overall UI focuses on:

- Minimal and modern design
- Eco-friendly green theme

- Easy navigation
- Mobile responsiveness
- Fast interaction
- User-friendly booking process

The interface is designed for:

- Students
- Daily commuters
- Tourists
- Admin staff with limited technical knowledge

3.2 Color Scheme

The system uses an eco-friendly green-based color palette to represent sustainability and electric transportation.

Purpose	Color
Primary Color	Green
Secondary Color	White
Accent Color	Dark Gray
Success Messages	Light Green
Error Messages	Red
Background	Light Gray/White

Design Justification

Green color is used because it represents:

- Clean energy
- Environment-friendly transportation
- Safety
- Trust

3.3 Typography

The system will use simple and readable fonts for better accessibility.

Element	Font Style
Headings	Bold Sans-serif
Body Text	Regular Sans-serif
Buttons	Medium-weight Sans-serif

Font Features

- Clear readability
- Mobile friendly
- Professional appearance

3.4 Responsiveness and Screen Constraints

The system is fully responsive and supports multiple screen sizes.

Supported Devices

- Laptops
- Mobile Phones

Responsive Features

- Flexible layouts
- Adaptive navigation menus
- Mobile-friendly buttons

Minimum Screen Resolution

- 320px width for mobile devices

Recommended Resolution

- 1920 × 1080 for desktop

3.5 Navigation Structure

The navigation system is simple and consistent throughout the application.

Main Navigation Menu

- Home
- Bikes
- How It Works
- Pricing

- About Us
- Contact
- Login/Register

User Dashboard Navigation

- Dashboard
- Book a Bike
- My Rides
- Payments
- Wallet
- Notifications
- Profile
- Settings
- Logout

Admin Dashboard Navigation

- Dashboard
- Bikes
- Users
- Bookings
- Payments
- Reports
- Settings
- Logout

3.6 User Interface Screens

3.6.1 Home Page

Purpose

The home page introduces the platform and encourages users to rent bikes.

Components

- Navigation bar
- Bike banner image

- “Rent a Bike” button
- Features section
- Footer

Information Displayed

- Available services
- Eco-friendly message
- Quick booking access
- Promotional content

Controls Used

Control	Purpose
Buttons	Navigation and booking
Navigation Links	Move between pages
Search Bar	Search bikes

3.6.2 Registration Page

Purpose

It allows new users to create an account.

Input Fields

- Full Name
- Email
- Phone Number
- Password
- Confirm Password

Controls

Control	Purpose
Textbox	Enter user information
Password Field	Secure password input
Register Button	Submit registration

Input Constraints

- Email must be valid

- Password minimum 8 characters
- Password must contain Alpha-Numeric values and at least one lower and upper case letter.
- Phone number numeric only

3.6.3 Login Page

Purpose

It allow the registered users only to access the system.

Input Fields

- Email
- Password

Features

- Forgot password option
- Secure login authentication

Controls

Control	Purpose
Email Field	Enter email
Password Field	Enter password
Login Button	Authenticate user

3.6.4 Bike Listing Page

Purpose

It displays available bikes for booking.

Information Displayed

- Bike image
- Bike model
- Battery percentage
- Price per hour
- Availability status

Features

- Search bikes
- Filter bikes

- Sort by popularity or price

Controls

Control	Purpose
Search Bar	Search bikes
Filter Panel	Filter bike types
Book Button	Reserve bike

3.6.5 Booking Page

Purpose

Allows users to reserve bikes and select rental duration.

Information Displayed

- Bike details
- Pickup time
- Return time
- Rental cost

Controls

Control	Purpose
Date Picker	Select booking date
Time Picker	Select rental duration
Continue Button	Proceed to payment

Input Constraints

- Pickup date cannot be past date
- Return time must be after pickup time

3.6.6 Payment Page

Purpose

It handles online payment processing.

Payment Methods

- JazzCash
- EasyPaisa

- Bank Transfer

Information Displayed

- Total fare
- Ride duration
- Payment summary

Controls

Control	Purpose
Payment Method Dropdown	Select payment method
Pay Now Button	Confirm payment

3.6.7 User Profile Management Page

Purpose

It allows users to manage their personal information and account settings. More detail are mentioned below:

Features

- Update profile
- Change password
- Upload profile picture
- View account summary

Information Displayed

- User_name
- Email
- Phone number
- Ride statistics

Controls

Control	Purpose
Text Fields	Edit personal data
Upload Button	Upload profile image
Save Changes Button	Update profile

Input Constraints

- Valid email format
- Image upload size limit
- Required fields validation

3.6.8 Ride History Page

Purpose

Displays previous rides and transaction history.

Information Displayed

- Ride date
- Duration
- Distance
- Cost

Controls

Control	Purpose
Table/Grid View	Display rides
Search Filter	Search ride records

3.6.9 Admin Dashboard

Purpose

It provides centralized management tools for admin and Staff.

Features

- Bike management
- User management
- Payment tracking
- Revenue reports
- Booking monitoring

Dashboard Components

- Statistics cards
- Charts and analytics
- Recent bookings table
- Revenue overview graph

Controls

Control	Purpose
Sidebar Menu	Navigate admin modules
Add Bike Button	Add new bikes
Edit/Delete Buttons	Manage records

3.7 Notifications and Alerts

The interface provides system notifications to improve user experience.

Notification Types

- Booking confirmation
- Payment success/failure
- Ride reminders
- Low battery alerts
- Error messages

Alert Styles

Alert Type	Color
Success	Green
Warning	Yellow
Error	Red

3.8 Security in User Interface

The UI includes several security-focused features.

Security Features

- Password masking
- Session timeout
- OTP verification
- Secure payment forms
- Role-based dashboard visibility

3.9 Future UI Enhancements

Future versions may include:

- Dark mode
- Urdu language support
- Voice assistance
- Real-time GPS map integration
- Mobile application interface
- AI-based personalized dashboard

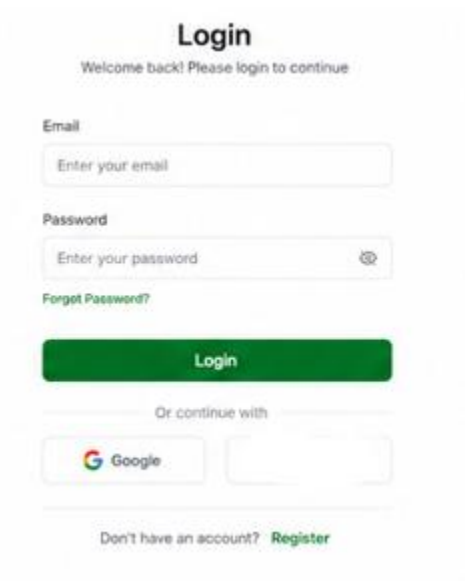
3.10 Reference Image of Frontend pages

In this section, Actual pages of website are include which provide more graphically acknowledgement of website front-end and user interaction with it.

3.10.1 Home Page Visual

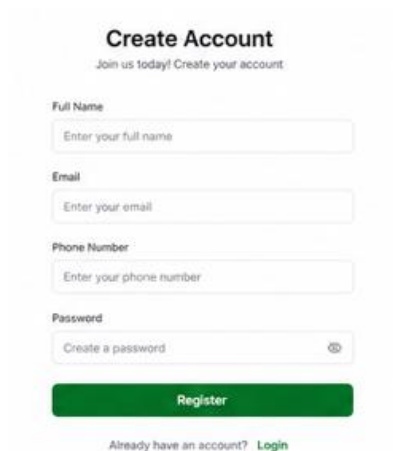


3.10.2 Login Page



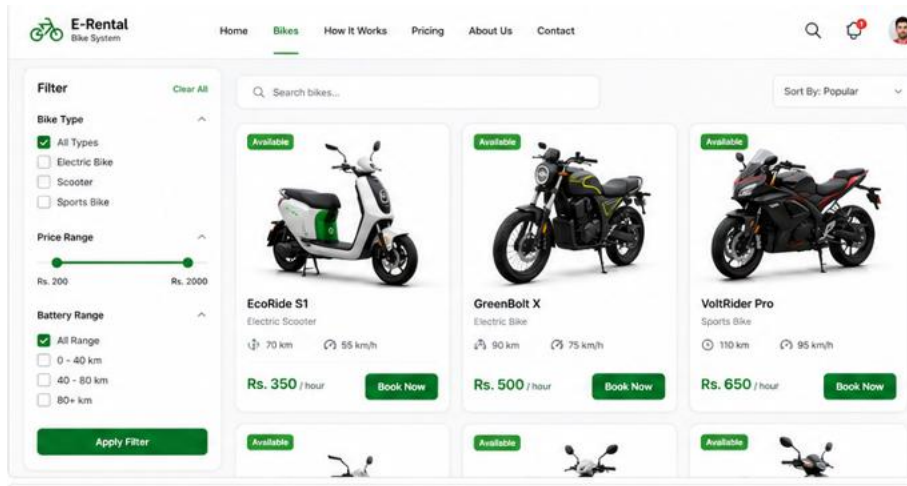
The login page features a central white card with a green border. At the top, the heading "Login" is displayed in bold black text, followed by the subtitle "Welcome back! Please login to continue" in a smaller font. Below this, there are two input fields: "Email" with the placeholder "Enter your email" and "Password" with the placeholder "Enter your password" and a toggle icon. A link "Forgot Password?" is positioned below the password field. A prominent green "Login" button is centered below the inputs. Underneath the button, the text "Or continue with" is followed by two social login options: a Google button with the Google logo and a Facebook button with the Facebook logo. At the bottom of the card, the text "Don't have an account? Register" includes a green "Register" link.

3.10.3 Register Page

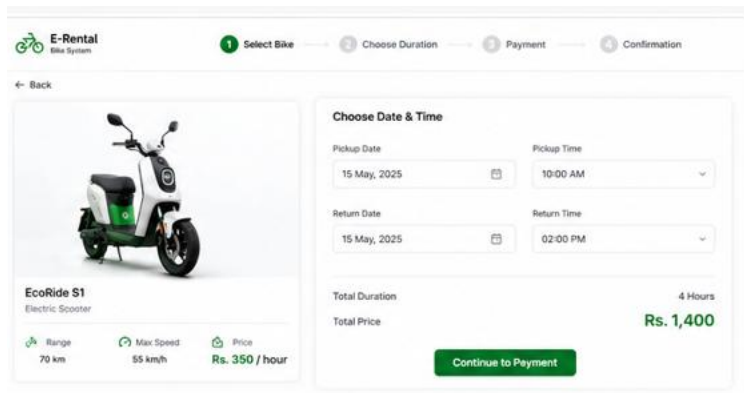


The registration page features a central white card with a green border. At the top, the heading "Create Account" is displayed in bold black text, followed by the subtitle "Join us today! Create your account" in a smaller font. Below this, there are four input fields: "Full Name" with the placeholder "Enter your full name", "Email" with the placeholder "Enter your email", "Phone Number" with the placeholder "Enter your phone number", and "Password" with the placeholder "Create a password" and a toggle icon. A prominent green "Register" button is centered below the inputs. At the bottom of the card, the text "Already have an account? Login" includes a green "Login" link.

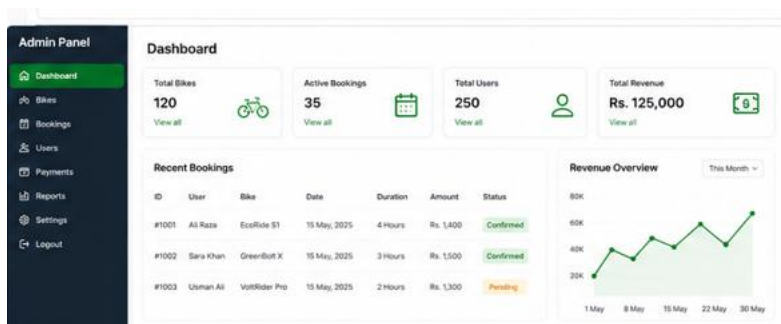
3.10.4 Bike Page



3.10.5 Bike Selection Page



3.10.6 Admin Dashboard Page



3.10.7 User Profile Management Page

